

paper, cardboard, textiles, plastic, wood, leather and ceramic, with little attenuation. Therefore it can be used for non-destructive and non-invasive sensing and imaging of targets under covers or in containers in some cases. Owing to these advantages, THz technologies are considered viable for inspecting hidden explosives, chemical threats and defects for defense and security applications.

THz spectroscopy and imaging of ERCs have been investigated intensively for possible defense and security applications. Measurements showing THz spectral features have been reported in a number of common energetic explosives, such as 2, 4, 6, trinitrotoluene (TNT), hexahydro-1, 3, 5-triazine (RDX), tetramethylene tetranitramine (HMX), pentaerythritol tetranitrate (PETN), and commercial explosives like PE-4 and Semtex. Most of these THz absorption fingerprints below 3 THz are from the lattice vibrational modes of solidstate explosive materials.

THz spectroscopy in transmission mode was used for explosives sensing in these investigations. However, for practical applications, reflection measurements are preferred since most bulky targets are impossible to measure in a transmission mode. Furthermore, reflection spectroscopy—specially diffused reflection spectroscopy—is more applicable for standoff detection than transmission methods. The schematic of the stand-off detection of explosives by reflection spectroscopy is shown in Fig. 5.

Compared to commonly used inspection techniques such as X-ray imaging, THz imaging offer advantages of spectroscopic information, non-ionising irradiation and higher image contrast for differentiating between various soft materials. These advantages stem from the fact that THz radiation is more sensitive to the nature of these materials as it passes through. For instance, an air bulb or a defect in the foam can be more easily identified with THz radiation rather

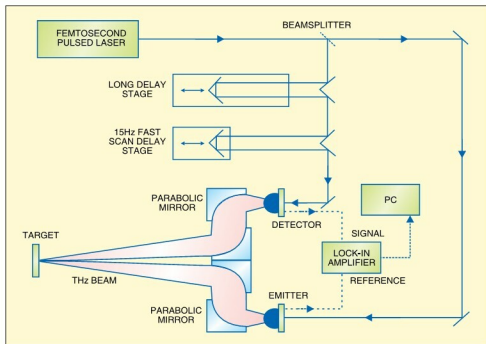


Fig. 5: Schematic of the standoff detection of explosives by reflection spectroscopy

Techniques for Generating THz Radiation

THz source/ electrical parameters	Optically pumped terahertz lasers	Time domain spectroscopy	Backward wave oscillators	Direct multiplied sources	Frequency mixing
Average power	>100 mW	~1 μ W	10 mW	mW to μ W (decreasing/ increasing frequency)	Tens of nW
Usable range	0.3-10 THz	0.1-2 THz	0.1-1.5 THz	0.1-1 THz	0.3-10 THz
Tunability	Discrete lines	NA	200 GHz	10-15 per cent of centre frequency	Continuous
CW/pulsed	CW or pulsed	Pulsed	CW	CW	CW

than X-ray in most cases.

Coherent THz radiation with an ultra-wide bandwidth can be used to characterise the electronic, vibronic and compositional properties of solids, liquids and gas phase materials.

In the case of dielectrics, the absorption is governed by optical phonons and depends on the polarity and optical phonon resonances of the material. On the other hand, metals are completely opaque to THz rays, and polar liquids such as water strongly absorb in this frequency region. In the gas phase, most polar molecules have very sharp and strong absorption lines in this frequency range, which reflects the unique rotational spectra of the absorbing species. Many chemical

components have distinct signatures responding to the THz wave due to molecular vibrations and rotational energy levels, implying that their chemical compositions might be examined using THz beam.

THz sources

Different techniques and sources of THz radiation are shown in the table.

In the THz regime, the best compact source technologies available currently produce only about 1 mW of average power—compared to the kilowatts or more commonly available at most other spectral frequencies. The lack of high-power, highly efficient and compact sources has been the greatest obstacle to exploiting THz.